125 mm (4.92 in) bore × 80 mm (3.15 in) rod Cylinder Force Data

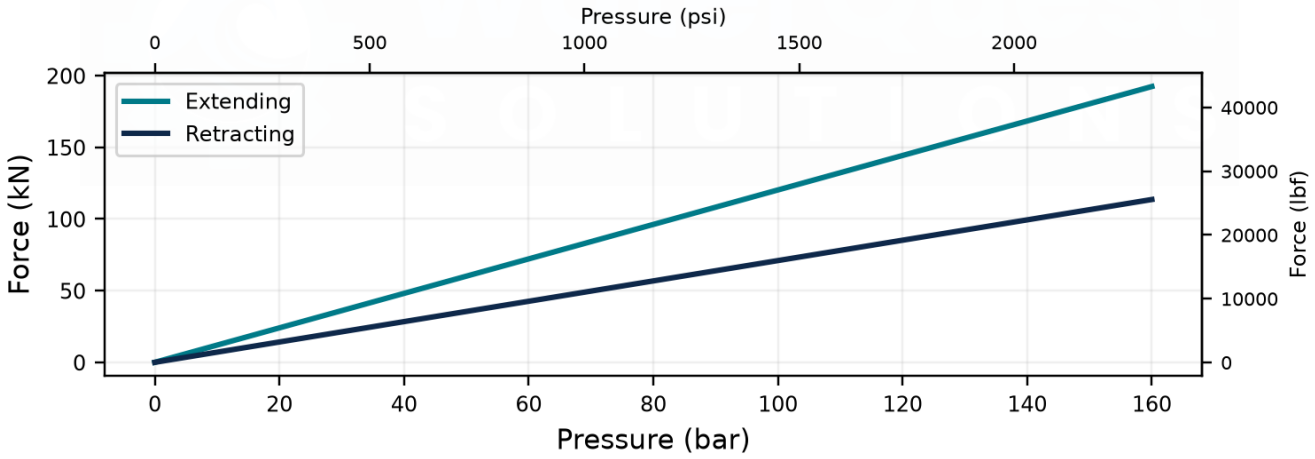
This geometry has a retraction area equal to 59.0% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 122.72 cm² (19.022 in²)	Ring Area 72.45 cm² (11.230 in²)
Max Extending Force 192.42 kN (43,258 lbf)	Max Retracting Force 113.60 kN (25,538 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	12.03 kN (2,704 lbf)	7.10 kN (1,596 lbf)
20 bar (290 psi)	24.05 kN (5,407 lbf)	14.20 kN (3,192 lbf)
30 bar (435 psi)	36.08 kN (8,111 lbf)	21.30 kN (4,788 lbf)
40 bar (580 psi)	48.11 kN (10,816 lbf)	28.40 kN (6,385 lbf)
50 bar (725 psi)	60.13 kN (13,518 lbf)	35.50 kN (7,981 lbf)
60 bar (870 psi)	72.16 kN (16,222 lbf)	42.60 kN (9,577 lbf)
70 bar (1,015 psi)	84.19 kN (18,927 lbf)	49.70 kN (11,173 lbf)
80 bar (1,160 psi)	96.21 kN (21,629 lbf)	56.80 kN (12,769 lbf)
90 bar (1,305 psi)	108.24 kN (24,333 lbf)	63.90 kN (14,365 lbf)
100 bar (1,450 psi)	120.27 kN (27,038 lbf)	71.00 kN (15,961 lbf)
110 bar (1,595 psi)	132.29 kN (29,740 lbf)	78.10 kN (17,558 lbf)
120 bar (1,740 psi)	144.32 kN (32,444 lbf)	85.20 kN (19,154 lbf)
130 bar (1,885 psi)	156.35 kN (35,149 lbf)	92.30 kN (20,750 lbf)
140 bar (2,031 psi)	168.37 kN (37,851 lbf)	99.40 kN (22,346 lbf)
150 bar (2,176 psi)	180.40 kN (40,556 lbf)	106.50 kN (23,942 lbf)
160 bar (2,321 psi)	192.42 kN (43,258 lbf)	113.60 kN (25,538 lbf)